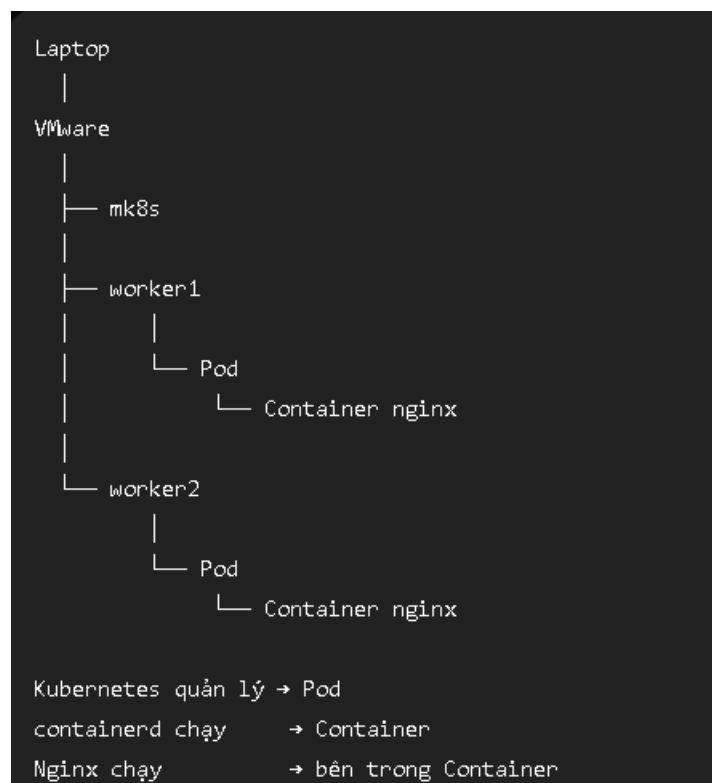


```
mk8s@mk8s:~$ kubectl get pods -l app=nginx-web -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP
nginx-web-776b46559d-2hkxr	1/1	Running	0	39s	10.233.105.131
nginx-web-776b46559d-f9nmd	1/1	Running	0	39s	10.233.125.6
nginx-web-776b46559d-pdfkg	1/1	Running	0	39s	10.233.105.132

Pod cơ bản



! hai container khác nhau nhưng nằm trong cùng Pod có thể giao tiếp qua localhost

Tạo pod thật

```
mk8s@mk8s:~$ vim pod-nginx.yaml
mk8s@mk8s:~$ kubectl apply --dry-run=client -f pod-nginx.yaml
pod/nginx-pod created (dry run)
mk8s@mk8s:~$ kubectl apply -f pod-nginx.yaml
pod/nginx-pod created
mk8s@mk8s:~$
```

```
mk8s@mk8s:~$ kubectl get pod nginx-pod -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NO
nginx-pod	1/1	Running	0	7m10s	10.233.125.7	worker2	<n

```
mk8s@mk8s:~$ kubectl get pod nginx-pod
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-pod	1/1	Running	0	8m15s

Xóa pod

```
mk8s@mk8s:~$ kubectl delete pod nginx-pod
pod "nginx-pod" deleted from default namespace
mk8s@mk8s:~$ kubectl get pod nginx-pod
Error from server (NotFound): pods "nginx-pod" not found
mk8s@mk8s:~$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-web-776b46559d-2hkxr	1/1	Running	0	27m
nginx-web-776b46559d-f9nmd	1/1	Running	0	27m
nginx-web-776b46559d-pdfkg	1/1	Running	0	27m

Multi-container Pod

tạo 1 pod chứa 2 container

Tạo manifest:

```
mk8s@mk8s:~$ vim multi-container-pod.yaml
mk8s@mk8s:~$ kubectl apply --dry-run=client -f multi-container-pod.yaml
pod/multi-container-pod created (dry run)
```

```

mk8s@m  worker1@  worker2@
apiVersion: v1
kind: Pod
metadata:
  name: multi-container-pod
  labels:
    app: multi-container-demo
spec:
  containers:
    - name: nginx
      image: nginx:1.25
      ports:
        - containerPort: 80
      resources:
        requests:
          memory: "32Mi"
          cpu: "25m"
        limits:
          memory: "128Mi"
          cpu: "200m"

    - name: busybox
      image: busybox:1.36
      command:
        - /bin/sh
        - -c
        - "while true; do sleep 3600; done"
      resources:
        requests:
          memory: "8Mi"
          cpu: "10m"
        limits:
          memory: "32Mi"
          cpu: "50m"

```

Tạo pod thật rồi check

```

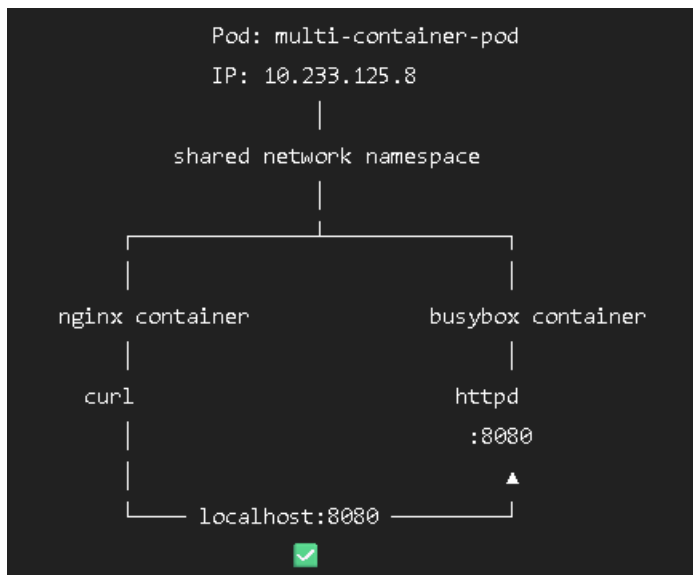
mk8s@mk8s:~$ kubectl apply -f multi-container-pod.yaml
pod/multi-container-pod created
mk8s@mk8s:~$ kubectl get pod multi-container-pod -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE
multi-container-pod                 2/2     Running   0           48s   10.233.125.8    work

```

```
mk8s@mk8s:~$ kubectl exec multi-container-pod -c nginx -- hostname -i
10.233.125.8
mk8s@mk8s:~$ kubectl exec multi-container-pod -c busybox -- hostname -i
10.233.125.8
```

Test Test Nginx → BusyBox

```
mk8s@mk8s:~$ kubectl exec multi-container-pod -c nginx -- \
  curl http://localhost:8080
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Curr
ent                                     Dload  Upload  Total  Spent    Left  Spee
d
  0         0     0     0     0     0      0      0  --:--:-- --:--:-- --:--:--
100    30  100    30     0     0 35587      0  --:--:-- --:--:-- --:--:-- 300
00
Hello from BusyBox container!
```

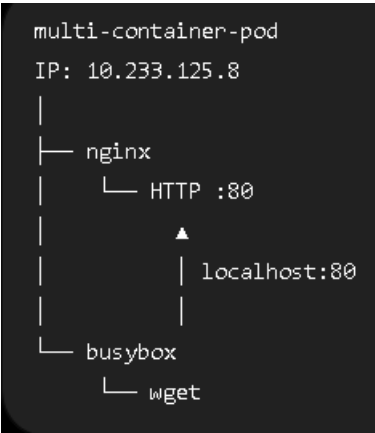


Test phía ngược lại

```
mk8s@mk8s:~$ kubectl exec multi-container-pod -c busybox -- \
  wget -qO- http://localhost:80
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```



ReplicaSet self-healing

Tạo 4 pods

```
mk8s@mk8s:~$ kubectl get pods --show-labels
```

NAME	READY	STATUS	RESTARTS	AGE	LABELS
db-dev	1/1	Running	0	6s	app=db,env=dev
db-prod	1/1	Running	0	3s	app=db,env=prod
multi-container-pod	2/2	Running	2 (77m ago)	19h	app=multi-container-demo,environment=lab
nginx-web-776b46559d-2hkxr	1/1	Running	1 (75m ago)	20h	app=nginx
nginx-web-776b46559d-f9nmd	1/1	Running	1 (77m ago)	20h	app=nginx
nginx-web-776b46559d-pdfkg	1/1	Running	1 (75m ago)	20h	app=nginx
web-dev	1/1	Running	0	12s	app=web,env=dev
web-prod	1/1	Running	0	9s	app=web,env=prod

Deploy tạo replica pods

```
mk8s@mk8s:~$ kubectl get pods -l app=web-lab -o wide
```

NAME	NODE	NOMINATED NODE	READY	STATUS	RESTARTS	AGE	IP
web-deployment-56c4f54f87-56zg6	125.17	worker2	1/1	Running	0	17s	10.233.125.17
web-deployment-56c4f54f87-g2jcv	125.16	worker2	1/1	Running	0	17s	10.233.125.16
web-deployment-56c4f54f87-v72z5	105.138	worker1	1/1	Running	0	17s	10.233.105.138

Test self healing

Kill 1 post để test

```
mk8s@mk8s:~$ kubectl get pods -l app=web-lab -o wide -w
```

NAME	NODE	NOMINATED NODE	READY	STATUS	RESTARTS	AGE	IP
web-deployment-56c4f54f87-8l9xq	3.125.19	worker2	1/1	Running	0	32s	10.233.125.19
web-deployment-56c4f54f87-g2jcv	3.125.16	worker2	1/1	Running	0	9m46s	10.233.125.16
web-deployment-56c4f54f87-m2zf7			1/1	Running	0	3m44s	10.233.125.17

```
mk8s@mk8s:~$ kubectl delete pod web-deployment-56c4f54f87-8l9xq
pod "web-deployment-56c4f54f87-8l9xq" deleted from default namespace
```

Tên và IP Pod đã thay đổi, nhưng ReplicaSet vẫn giữ nguyên

web-deployment-56c4f54f87-8l9xq	1/1	Terminating	0	48s	1
0.233.125.19 worker2 <none>		<none>			
web-deployment-56c4f54f87-jw5c5	0/1	Pending	0	0s	<
none> <none> <none>		<none>			
web-deployment-56c4f54f87-8l9xq	1/1	Terminating	0	48s	1
0.233.125.19 worker2 <none>		<none>			
web-deployment-56c4f54f87-jw5c5	0/1	Pending	0	0s	<
none> worker2 <none>		<none>			
web-deployment-56c4f54f87-jw5c5	0/1	ContainerCreating	0	0s	
<none> worker2 <none>		<none>			
web-deployment-56c4f54f87-8l9xq	1/1	Terminating	0	48s	
10.233.125.19 worker2 <none>		<none>			
web-deployment-56c4f54f87-8l9xq	0/1	Completed	0	48s	
10.233.125.19 worker2 <none>		<none>			
web-deployment-56c4f54f87-jw5c5	0/1	ContainerCreating	0	0s	
<none> worker2 <none>		<none>			
web-deployment-56c4f54f87-jw5c5	0/1	ContainerCreating	0	1s	
<none> worker2 <none>		<none>			
web-deployment-56c4f54f87-jw5c5	1/1	Running	0	1s	
10.233.125.20 worker2 <none>		<none>			
web-deployment-56c4f54f87-8l9xq	0/1	Completed	0	49s	
10.233.125.19 worker2 <none>		<none>			
web-deployment-56c4f54f87-8l9xq	0/1	Completed	0	49s	
10.233.125.19 worker2 <none>		<none>			

```

mk8s@mk8s:~$ kubectl get deployment web-deployment
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
web-deployment 3/3     3            3           14m
mk8s@mk8s:~$ kubectl get rs -l app=web-lab
NAME                                DESIRED   CURRENT   READY   AGE
web-deployment-56c4f54f87          3         3         3       14m

```

```

mk8s@mk8s:~$ kubectl get pods -l app=web-lab -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP
web-deployment-56c4f54f87-g2jcv     1/1    Running   0          11m   10.23
3.125.16 worker2 <none>              <none>
web-deployment-56c4f54f87-jw5c5     1/1    Running   0          110s  10.23
3.125.20 worker2 <none>              <none>
web-deployment-56c4f54f87-m2zz7     1/1    Running   0          5m50s 10.23
3.105.139 worker1 <none>             <none>

```

bỏ Deployment ra và **tự tạo ReplicaSet** trực tiếp:

tùy chỉnh scale pod

từ 3-5 pods

```

mk8s@mk8s:~$ kubectl scale rs rs-nginx-lab --replicas=5
replicaset.apps/rs-nginx-lab scaled
mk8s@mk8s:~$ kubectl get pods -l app=rs-nginx -w
NAME                                READY    STATUS    RESTARTS   AGE
rs-nginx-lab-h8xf4                 1/1      Running   0           5s
rs-nginx-lab-pf749                 1/1      Running   0           5s
rs-nginx-lab-qgkkt                 1/1      Running   0          3m14s
rs-nginx-lab-tcj6x                 1/1      Running   0          3m14s
rs-nginx-lab-w65c5                 1/1      Running   0          3m14s
^Cmk8s@mk8s:~$ kubectl get rs rs-nginx-lab
NAME            DESIRED    CURRENT    READY    AGE
rs-nginx-lab    5          5          5        3m30s

```

Từ 5 → 2 pods

```

mk8s@mk8s:~$ kubectl scale rs rs-nginx-lab --replicas=2
replicaset.apps/rs-nginx-lab scaled
mk8s@mk8s:~$ kubectl get pods -l app=rs-nginx -w
NAME                                READY    STATUS    RESTARTS   AGE
rs-nginx-lab-qgkkt                 1/1      Running   0          3m45s
rs-nginx-lab-w65c5                 1/1      Running   0          3m45s
^Cmk8s@mk8s:~$ kubectl get rs rs-nginx-lab
NAME            DESIRED    CURRENT    READY    AGE
rs-nginx-lab    2          2          2        3m54s

```

Xóa pod test replicas

```

mk8s@mk8s:~$ kubectl get pods -l app=rs-nginx
NAME                                READY    STATUS    RESTARTS   AGE
rs-nginx-lab-qgkkt                 1/1      Running   0          6m42s
rs-nginx-lab-w65c5                 1/1      Running   0          6m42s
mk8s@mk8s:~$ kubectl delete pod rs-nginx-lab-qgkkt
pod "rs-nginx-lab-qgkkt" deleted from default namespace
mk8s@mk8s:~$ |

mk8s@mk8s:~$ kubectl get pods -l app=rs-nginx
NAME                                READY    STATUS    RESTARTS   AGE
rs-nginx-lab-5fjv6                 1/1      Running   0          28s
rs-nginx-lab-w65c5                 1/1      Running   0          7m34s

```

NAME	READY	STATUS	RESTARTS	AGE	IP	N
rs-nginx-lab-qgkkt	1/1	Running	0	6m32s	10.233.125.21	w
orker2 <none>		<none>				
rs-nginx-lab-w65c5	1/1	Running	0	6m32s	10.233.105.140	w
orker1 <none>		<none>				
rs-nginx-lab-qgkkt	1/1	Terminating	0	7m6s	10.233.125.21	
worker2 <none>		<none>				
rs-nginx-lab-5fjv6	0/1	Pending	0	0s	<none>	
<none> <none>		<none>				
rs-nginx-lab-qgkkt	1/1	Terminating	0	7m6s	10.233.125.21	
worker2 <none>		<none>				
rs-nginx-lab-5fjv6	0/1	Pending	0	0s	<none>	
worker2 <none>		<none>				
rs-nginx-lab-5fjv6	0/1	ContainerCreating	0	0s	<none>	
worker2 <none>		<none>				
rs-nginx-lab-qgkkt	1/1	Terminating	0	7m6s	10.233.1	
25.21 worker2 <none>		<none>				
rs-nginx-lab-qgkkt	0/1	Completed	0	7m6s	10.233.1	
25.21 worker2 <none>		<none>				
rs-nginx-lab-5fjv6	0/1	ContainerCreating	0	1s	<none>	
worker2 <none>		<none>				
rs-nginx-lab-5fjv6	0/1	ContainerCreating	0	1s	<none>	
worker2 <none>		<none>				
rs-nginx-lab-5fjv6	1/1	Running	0	1s	10.233.1	
25.23 worker2 <none>		<none>				
rs-nginx-lab-qgkkt	0/1	Completed	0	7m7s	10.233.1	
25.21 worker2 <none>		<none>				
rs-nginx-lab-qgkkt	0/1	Completed	0	7m7s	10.233.1	
25.21 worker2 <none>		<none>				
rs-nginx-lab-5fjv6	1/1	Running	0	81s	10.233.1	
25.22 worker2 <none>		<none>				

Xóa cả 2 pod cùng lúc

```
mk8s@mk8s:~$ kubectl get pods -l app=rs-nginx
NAME                READY   STATUS    RESTARTS   AGE
rs-nginx-lab-5fjv6   1/1     Running   0           28s
rs-nginx-lab-w65c5   1/1     Running   0           7m34s

mk8s@mk8s:~$ kubectl delete pod -l app=rs-nginx
pod "rs-nginx-lab-5fjv6" deleted from default namespace
pod "rs-nginx-lab-w65c5" deleted from default namespace
```

Vẫn replican về 2 máy

```
mk8s@mk8s:~$ kubectl get pods -l app=rs-nginx
NAME                READY   STATUS    RESTARTS   AGE
rs-nginx-lab-s8qmq   1/1     Running   0           101s
rs-nginx-lab-t8l6g   1/1     Running   0           101s
```